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Evaluating Manipulative Language Taxonomies for Human and LLM-Based Classification of Political Advertisements

Final Project Report

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Abstract

This report evaluates the suitability of three popular manipulative-language taxonomies for both human annotation and LLM-based classification of political advertisements. Although prior research has applied manipulation taxonomies in manipulative language classification tasks, little attention has been given to whether the taxonomies themselves are well suited to support consistent annotation or automated detection. This project addresses that gap through a comparative evaluation of three established manipulation taxonomies: Noggle's taxonomy, Simon's taxonomy, and Information Manipulation Theory 2 (IMT2).

Using a dataset of 600 advertisements taken from the 'Comparable 2022 General Election Advertising Datasets from Meta and Google' dataset, the taxonomies were evaluated through human annotation, zero-shot and few-shot LLM classification, and the taxonomy quality metrics established by Unterkalmsteiner and Abdeen.

Human annotators showed a preference for Noggle's taxonomy, which achieved the strongest comprehensiveness and intra-model reliability. In contrast, IMT2 yielded the strongest LLM classification performance, despite weak agreement across models. Few-shot prompting did not consistently improve results.

The findings reveal a trade-off between human interpretability and model performance, suggesting that taxonomies better suited to human reasoning may not necessarily be successfully implemented by LLMs. Alongside this, the results indicate that taxonomy design itself influences classification outcomes and should be treated as an object of evaluation rather than assumed as a fixed foundation for manipulation detection.

Originality Avowal

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Jasmine Lever

22 April 2026

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Chapter 1

Introduction

Manipulative language in communications has existed for as long as humans have had the ability to persuade, and advances in technology have only given us better channels through which to do so[1]. However, with the rise of personalised online content, targeted advertising now has unprecedented potential to utilise manipulative language with significant impacts. This is particularly concerning in the political sphere, with potential knock-on effects on voting patterns, misinformation spread and political polarisation. Despite its importance, it is not straightforward to determine how to reliably classify manipulative language in political adverts.

Consider this advert from *Focus On The Family*, an American Evangelical Protestant organisation.

"An ultrasound can change the course of life! When a future mom sees her little one for the first time, she'll fall in love - and choose LIFE."

A survey asking whether this advert is "manipulative" might yield a majority answer of *yes*. However, respondents might disagree entirely about *why*. One person may identify the message as "shaming"; another might call it "highly emotive". With so many different understandings of the same concept, it becomes subjective and analytically unhelpful. In classification, this becomes a methodological challenge, as before any classifier model can be trained, researchers must first decide what labels ought to be assigned in the first place. Taxonomies attempt to resolve this issue, providing structured frameworks to make annotations more consistent and well-defined.

In the study of manipulative language, three taxonomies dominate the field: those of Noggle [2], Simon [3]and McCornack [4]. These frameworks may offer more rigour than unstructured intuition, but they have never been compared systematically. Whilst there is research applying these taxonomies to the classification of manipulation in text [5] [6] [7], no research has been carried out on the quality of the taxonomies themselves for those tasks. Taxonomy evaluation is an emerging research area that questions the assumption that the existence of a framework implies its adequacy. A taxonomy may be widely used but still fails to capture the conceptual space it claims to describe.

To support empirical analysis, this paper utilises the large-scale dataset 'Comparable 2022 General Election Advertising Datasets from Meta and Google'[8] of political advertisements from the 2022 U.S. general election, compiled from Meta Platforms and Google transparency archives. The dataset includes textual and multimodal campaign content alongside metadata, making it well-suited for evaluating how manipulation taxonomies perform on real-world political messaging.

The goal of this paper is to evaluate the utility of these three manipulation taxonomies at two levels using this dataset. The contributions are the following:

1. **Human Annotation:** Provide human annotators with the taxonomy's rules and ask them to label 600 political adverts accordingly.
2. **Model Evaluation:** Implement the taxonomy rules using multiple large language model (LLM) classifiers, specifically GPT, Gemini, and Claude. Their performance will be assessed using the metrics F1 score, recall, and precision.
3. **Taxonomy Quality Metrics**: Using both the human annotations and model outputs, we then compute several taxonomy quality metrics outlined by Unterkalmsteiner and Abdeen[9]. These will help us identify structural limitations within each taxonomy.

The following research questions guide this paper:

1. To what extent do the different manipulation taxonomies align with human perceptions of manipulation in political advertisements?
2. How does taxonomy choice affect LLM performance in manipulation classification?
3. How consistent are predictions across models for each taxonomy?
4. How do the taxonomies compare in terms of structural quality, according to established taxonomy-evaluation metrics?

Chapter 2

Literature Review

2.1 Manipulative Language in Political Advertisements

Van Dijk's foundational work on manipulation[10] describes its "fuzzy" nature, particularly in the difficulty of distinguishing legitimate persuasion from illegitimate manipulation. Van Dijk pinpoints that the crucial distinction lies in their consequences - *manipulation* specifically occurs when the recipient is "unable to understand the real intentions or to see the full consequences of the beliefs or actions advocated by the manipulator". The paper further asserts that manipulation follows two criteria: first, that the recipient is acted upon against their full conscious will or interests, and second, that the manipulation primarily serves the interests of the manipulator.

As a result of these criteria, van Dijk characterises manipulation as normatively wrong, not just because it violates conversational norms, but because it is illegitimate within a democratic society. The paper further highlights how manipulation typically reproduces social inequality, benefiting powerful speakers whilst harming the less powerful recipients. The problem of manipulation then, is not just a communicative one, but a political one.

The problem becomes even more significant in the context of political advertising, a high-stakes environment where the consequences extend beyond individuals; manipulation in this domain has direct influence on voting outcomes, polarisation and the shape of public discourse. As Aliyeva[11], drawing on Kilby[12], argues, media functions as a "magnifying glass" that transmits ideas, doctrines, and information to society at large. In digital contexts, this magnifying effect is intensified by the one-to-many nature of advertising; recipients lack any meaningful opportunities to respond, challenge, or interrogate messages in the same communicative space.

This reinforces the "passivity" that recipients of manipulation adopt according to van Dijk[10], a structural asymmetry in communicative power that benefits the manipulator. The rise of digital platforms further contributes to these dynamics. As Suwarni notes, political campaigns are now able to "deploy tailored language that aligns with specific ideological, demographic and temporal contexts"[13] through micro-targeting and personalised messaging.

This project builds on these theoretical foundations by addressing a key gap in the literature: the lack of systematic evaluation of whether normative and discourse-theoretic accounts of manipulation can be translated into reliable annotation frameworks for both human annotators and automated systems.

2.2 Manipulative Language Taxonomies

This paper explores three key taxonomies of manipulation that dominate research about machine-learning manipulation detection.

2.2.1 Noggle's Taxonomy

Noggle's taxonomy is defined in the chapter "Thirteen Forms of Manipulation"[2]. Using examples both "real [or] psychologically realistic", the chapter lists the following as forms of manipulation:

1. Playing on the Emotions
2. Misdirection / Misemphasis
3. Gaslighting
4. Game Switching
5. Social Proof
6. Peer Pressure
7. Emotional Blackmail
8. Nagging
9. Charm Offensive
10. Exploiting Conflict Aversion

11. Exploiting Reciprocity
12. Negging
13. Conditional Flattery

It is important to note, however, that the chapter lays out these 13 ”paradigm examples” with the purpose of serving as a dataset against which nine different accounts of manipulation will be assessed. In other words, the chapter sets out to describe what manipulation looks like, in order for it to evaluate what the best definition of manipulation is and what it is rooted in. The area of the book we specifically utilise is the thirteen forms, however it is necessary to contextualise the task Noggle was overall trying to achieve. Indeed, Noggle writes that the intention of the chapter specifically is to “document [the] diversity”[2] of manipulation; he even acknowledges that the list may not be necessarily complete, but at least sufficiently long enough to serve its purpose as a dataset.

2.2.2 Simon’s Taxonomy

Simon’s taxonomy[3] lays out nineteen different manipulative tactics:

1. Minimisation
2. Lying by omission
3. Lying by distortion
4. Denial
5. Selective Inattention
6. Rationalisation
7. Diversion
8. Evasion
9. Covert intimidation
10. Guilt-tripping
11. Shaming
12. Playing the Victim Role

13. Vilifying the victim

14. Playing the servant role

15. Seduction

16. Projecting the blame

17. Feigning innocence

18. Feigning ignorance or confusion

19. Brandishing anger

It is important once again to note the purpose of the text: Simon explicitly states that the book was written "for the general public as well as the mental health professional"[3], after noticing that many of the symptoms of depression and anxiety experienced by his patients could be attributed to their relationships with manipulative individuals. As such, the taxonomy's core purpose is to be used in personal settings, such as applied to one's own life, or in "therapeutic interventions". Simon asserts that the taxonomy is highly accurate, more so than previous descriptions of manipulation that were proposed before the cognitive-behavioural approach to psychology was fully fledged.

2.2.3 IMT2

IMT2, or Information Manipulation Theory 2, is a propositional theory rooted directly in artificial intelligence as well as cognitive neuroscience and psychology[4] IMT2 attempts to expand on McCormack's original theory which had initially been criticised for not being testable. IMT2 questions the dichotomous model of BFL (bald-faced lies) and BFT (bald-faced truths) previously presumed in deception research and instead proposes that deceptive and truthful statements are both produced by an incremental construction process. Alongside this central premise, McCornack defines three propositional sets, the most important for this paper being the Information Manipulation set. Inherited from Grice's maxims[14], it serves to explain "how exactly information is manipulated within spoken discourse". The four maxims are as follows:

1. Quantity - "the amount of relevant information that is shared"

2. Quality - "the veracity of shared information"

3. Manner - "The way in which disclosed information is expressed"

4. Relation - "The relevance of disclosed information"

An important aspect of the theory, particularly in relation to the research presented in this paper, is its grounding in concepts taken from scholarship in artificial intelligence: in particular, the theory adopts the notion of opportunistic problem solving to model deceptive discourse as an incremental planning process, helping to explain how deception can arise "mid-utterance" without prior intent to deceive. It should be noted however, that this grounding in artificial intelligence serves to model cognitive systems more informatively, rather than to propose a framework for direct implementation in machine learning systems.

It is clear that these taxonomies emerge from distinct theoretical and disciplinary backgrounds. While they were originally developed for philosophical, psychological, or discourse-analytic purposes, this paper evaluates how effectively each can be repurposed for the task of automated manipulation detection, assessing their suitability for both human annotation and LLM-based classification.

2.3 Machine-Learning Approaches to Manipulative Language Classification

Across the literature, it is highlighted that the task of manipulative language classification is more difficult to implement successfully than that of similar tasks such as toxicity[6] or misinformation[15] detection. The paper "Detecting Manipulative Narratives in Social Media"[15] stresses this, identifying that the core challenge in detection is identifying "the exact strategies being used in a given piece of text". In "MentalManip: A Dataset For Fine-grained Analysis of Mental Manipulation in Conversations"[6], Wang et al attribute this to the fact that manipulation is more context-dependent, requiring interpretation of implicit speaker intent.

The SemEval[16] paper acknowledges the distinction between two tasks of differing difficulty: determining whether a text is manipulative, and identifying the specific manipulative strategies being used. They note that the results of 250 teams attempting to do both subtasks show that the subtask technique classification "proved to be much more challenging" than span identification. It is noted that some teams were unable to outperform the logistic regression baseline that classified using only the length of the fragment. The highest F1 score achieved was 0.5155 in testing and 0.5012 in development.

These findings highlight a key limitation in current approaches: while the difficulty of fine-grained manipulation classification is well established, there has been relatively little focus on whether this difficulty stems from the modelling approaches used, or from the underlying taxonomies that define the classification task itself. In other words, existing work largely assumes that the categories of manipulation are well-defined and suitable for annotation, and instead concentrates on improving model performance.

This paper addresses this gap by shifting the focus from model optimisation to taxonomy evaluation. By applying multiple manipulation taxonomies to the same dataset and comparing their performance across both human annotation and LLM-based classification, this study investigates whether differences in classification difficulty can be attributed, at least in part, to the structure and clarity of the taxonomies themselves.

2.4 Taxonomy Adoption as a Design Choice

Across the literature, manipulation taxonomies are typically adopted rather than interrogated. ChatbotManip[17]is explicit about this limitation, stating that they do not claim that the tactics used are "a comprehensive list of ways in which a chatbot may manipulate a human"[17]. They justify implementing Noggle's taxonomy due to its perceived "sufficient granularity", as well as making the claim that the categories described in it are more common to everyday life than the categories described within Simon's taxonomy, making it more useful for general public annotators. Thus while some consideration has gone into the choice of taxonomy, it is evidently subjective rather than mathematically grounded.

In "MentalManip: A Dataset For Fine-grained Analysis of Mental Manipulation in Conversations"[6], a multi-level taxonomy is constructed, inspired by Simon's work. They acknowledge that the category definitions evolved through expert consultation and annotator feedback during dataset construction. This reflects a broader pattern: taxonomies are treated as working frameworks shaped by practical constraints rather than theoretical modelling.

This has direct implications for the objectives of this paper: if manipulation taxonomies are typically adopted without systematic evaluation, then their suitability for tasks such as automated manipulation detection cannot be assumed. Accordingly, this paper treats taxonomy selection itself as an object of evaluation, comparing multiple frameworks to determine how their underlying structures affect annotation consistency and model performance.

2.5 Annotation Methodologies

Annotation practices across manipulation datasets range greatly, but agreement remains moderate. ChatbotManip[17] uses a crowdsourcing protocol in which annotators must achieve "100% accuracy in a screening survey" before participating. The labels are collected on a "7-point Likert scale," and inter-annotator agreement is reported. Despite these efforts, agreement dropped significantly when annotators were asked to distinguish between manipulation types: Krippendorff's alpha fell from **0.61** (general manipulation) to **0.49** (all classes). This reflects the aforementioned contrast in difficulty in the two tasks of detection and classification.

MentalManip[6] adopts a more structured protocol, assigning three annotators per dialogue, avoiding repeated annotator pairings, and allowing annotators to select "cannot decide" when unsure. The dataset was then composed of over 4000 high-confidence dialogues after aggregation. Despite this, the Fleiss' Kappa score was found to be **0.596**, what the authors describe as "moderate annotator agreement,". They explicitly credit this to the fact that "the judgment of manipulation is very subjective". The paper "Crowdsourced Detection of Emotionally Manipulative Language"[18] further demonstrates that annotation errors may be systemic rather than random, as they highlight that annotations frequently "struggle to disentangle" emotionally manipulative language and emotional intensity, what they call "conflation error".

These findings have direct implications for this paper. Given that annotator agreement remains only at a "moderate" level, it is necessary to consider whether this difficulty arises solely from the subjective nature of manipulation, or from the way it is defined. This paper therefore treats annotation as a diagnostic tool, using both annotator agreement and model performance to evaluate the practical utility of the three manipulation taxonomies under consideration.

2.6 Large Language Models and Prompt-Based Classification

For the purpose of evaluating the applicability of the three taxonomies, this paper employs large language models (LLMs) as manipulation classifiers under both zero-shot and few-shot prompting conditions. Specifically, zero-shot classification is conducted using GPT-4o-mini, Gemini, and Claude, while few-shot prompting is additionally applied to GPT-4o-mini to assess whether limited examples improve alignment with taxonomy definitions.

The use of zero-shot and few-shot LLMs for manipulation detection is grounded in recent work, most notably ChatbotManip[17], where multiple models were evaluated under similar conditions for identifying manipulative language in chatbot conversations. Their findings indicate that model scale does not significantly impact performance, with only "a small difference between zero-shot, few-shot and chain-of-thought" prompting. Although the Gemini model achieved the strongest overall results, no configuration was considered suitable for real-world deployment due to high rates of false positives.

MentalManip[6] reports similar results, with high false positive rates and substantial variation across models. While GPT-4 Turbo achieved an accuracy of 0.653 on 899 dialogues, smaller LLaMA-based models performed significantly worse, with accuracies of 0.004 and 0.022. The authors attribute this in part to the difficulty of aligning model predictions with subjective human judgments, noting that "the subjectivity of human judgment complicates the alignment of AI models' predictions with human choices".

In this context, the use of LLM prompting in this paper is not motivated by claims of superior detection performance. Instead, LLMs are used as a tool for evaluating the taxonomies themselves. Comparing the model-generated labels with human annotations enables us to evaluate the extent to which each taxonomy supports consistent interpretation across both humans and LLMs, and to identify which framework produces the strongest alignment between the two.

2.7 Taxonomy Evaluation

Despite the central role that taxonomies play in structuring complex domains, there is broad agreement that taxonomy evaluation remains underdeveloped and inconsistently applied. Seeking to extend Nickerson et al's[19] work taxonomy development method, Szopinski et al.[20] provide an overview of the criteria most commonly used to evaluate taxonomies. They identify 43 different evaluation criteria from their sample of articles and propose guidance on how to employ them. From this, they organise evaluation criteria around three dimensions: ending conditions, usefulness and applicability, of which, the last two are most applicable to this paper. Usefulness describes evaluating whether applying the taxonomy improves task performance; applicability describes the feasibility of applying the taxonomy to real entities in the domain. They observed that a majority of articles merely state these qualities rather than empirically demonstrate them. In their guidance, Szopinski et al.[20] suggest evaluating for these characteristics with a case study attempting classification for example (if that is the purpose of the taxonomy), and comparing the results of two groups of participants: those who attempted to classify with the taxonomy, and those who did not have access to the taxonomy.

Kaplan et al.[21] similarly propose a method for Taxonomy Evaluation, following the Goal-Question-Metric approach. The method consists of three steps: evaluating structural suitability, applicability and purpose. Their evaluation method is grounded mathematically; step one verifies generality and appropriateness through the measures of laconicity, lucidity, completeness and soundness. The other two steps, conversely, draw parallels to the notions of usefulness and applicability seen above. The paper similarly suggests examining the taxonomy in use by participants, however they focus more on whether it can be used consistently. This is assessed through inter-annotator agreement, as well as comparison to a gold standard.

Finally, we consider the article 'A compendium and evaluation of taxonomy quality attributes'[9]. It aims to "identify taxonomy quality attributes that provide practical measurements". The work reanalyses Szopinski et al.'s 43 listed criteria[20] through the lens of measurement theory; the paper argues that usefulness and applicability are inherently subjective, and focuses instead on internally measurable taxonomy attributes. They map seven such attributes to concrete measurements. From these, we adopt a subset suited to our evaluation setting. Specifically, we evaluate comprehensiveness via the count of unclassified objects; robustness via the number of taxonomy constructs and semantic proximity analysis using word embeddings; conciseness via unused constructs and human memory heuristics; mutual exclusiveness via ambiguity in classifications; and reliability through inter- and intra-rater agreement.

In this study, reliability is assessed across three dimensions in order to capture the consistency of taxonomy application in both human and automated settings. First, inter- and intra-annotator agreement is used to evaluate the extent to which human annotators consistently apply the taxonomy. Second, agreement between human annotations and LLM-generated classifications is used to assess how well each taxonomy translates into a form that can be interpreted by automated systems in alignment with human understanding. Third, inter- and intra-model agreement is examined to determine whether LLMs apply the taxonomy consistently across repeated classifications and across different models. Together, these measures allow us to evaluate not only the internal coherence of each taxonomy, but also its reliability as a shared framework between human and machine interpretation.

Chapter 3

Methodology

3.1 Design Overview

This study evaluated three manipulative-language taxonomies as classification schemes for political advertisements, and tested how well each taxonomy could be employed by (i) human annotators, (ii) a large language model prompted with taxonomy-specific instructions. The evaluation consisted of two components: taxonomy-quality metrics, which assessed how consistently and coherently each taxonomy could be applied by human annotators, and model performance metrics, which assessed how well each taxonomy supported accurate prediction when used by LLMs. Together, these allowed us to evaluate both the human interpretability of each taxonomy and the applicability to automated classification.

3.2 Human annotation design

Annotation was conducted in three independent rounds, one per taxonomy. In each round:

- Annotators received only the relevant taxonomy's category definitions and decision rules.
- Annotators were not shown their own earlier labels or others' labels.

A subset of 75 advertisements was annotated by multiple annotators; each annotator labelled 5 adverts that were also part of another annotator's subset. These overlapping advertisements were selected randomly. This approach balanced the need for reliable agreement measurement with the practical constraints of annotation workload, ensuring that sufficient overlap existed for evaluation while maintaining feasibility for participants.

Inter-annotator agreement was computed separately for each taxonomy using Krippendorff's α [22] and Gwet's AC1 [23], enabling direct comparison of how consistently each taxonomy could be applied.

3.3 LLM-based classification

Three separate LLM classification pipelines were constructed, one for each taxonomy. For each taxonomy, zero-shot classification was performed using GPT, Claude, and Gemini, where models were prompted with taxonomy-specific instructions but were provided with no labelled examples.

In addition, few-shot classification was conducted using GPT only, incorporating 12 annotated examples per taxonomy. These examples were selected from instances with the highest level of agreement between annotators wherever possible, in order to provide clear and consistent guidance to the model. Model parameters were held constant across taxonomies, and the specific model versions and configurations were recorded.

This approach evaluated whether each taxonomy could be utilised through natural-language instructions, testing whether its conceptual categories were sufficiently clear and separable to be applied consistently by a prompted model.

3.4 Evaluation Framework

3.4.1 Model performance metrics

For both automated pipelines, predictions were evaluated against the human annotated set for each taxonomy independently. The following metrics were computed:

1. Micro-averaged precision
2. Recall

3. F1 score
4. Macro-averaged F1 score
5. Per-class precision, recall and F1 score

These metrics provided insight into how usable each taxonomy was when implemented in automated systems - targeting the “usefulness” criteria described consistently across taxonomy evaluation literature.

3.4.2 Taxonomy Quality Evaluation

Taxonomy quality was evaluated following the measurement framework proposed by Unterkalmsteiner and Abdeen[9], which evaluated taxonomy quality in terms of external attributes assessed through observable measurements. The following subset of measurements was chosen for this paper due to feasibility:

Comprehensiveness:

- Count of unclassified objects

Robustness:

- Semantic proximity and distance between taxonomy constructs, measured using Sentence-BERT[24] embeddings to assess similarity between category definitions.

Conciseness:

- Count of unused constructs: categories or characteristics never used in practice, based on annotator usage.

Mutual Exclusiveness:

- Analysis of label co-occurrence patterns to identify overlap between categories.

Reliability:

- Inter-annotator reliability: agreement between different human annotators.
- Intra-model reliability: Comparing repeated outputs from the same LLM under identical conditions. This is a proxy to intra-annotator reliability under feasibility constraints.

Chapter 4

Experimental Setup

4.1 Dataset Setup

From the larger 'Comparable 2022 General Election Advertising Datasets from Meta and Google'[8] dataset, a subset of 600 Meta advertisements were selected. Only advertisements from the Meta (Facebook) Ad Library were included in this study in order to ensure consistency in data formatting and structure. The Meta and Google components of the original dataset differ in how advertisement content and metadata are represented, which would introduce additional variability unrelated to the evaluation of the taxonomies. Restricting the dataset to a single platform allows for a more controlled and consistent annotation and evaluation process.

The subset of 600 advertisements was randomly sampled from the dataset following preprocessing. Textual content from multiple fields (ad body text, captions, and OCR/ASR outputs) was combined into a single representation for each advertisement.

Advertisements with empty text were removed, and duplicate entries were filtered based on *ad_id* to ensure each instance was unique. From the resulting dataset, 600 advertisements were sampled without replacement using a fixed random seed to ensure reproducibility.

4.2 Human Annotation Procedure

There were 16 annotators in total. Each annotator labelled either 36 or 42 advertisements (corresponding to 6% or 7% of the constructed dataset). Each advertisement was annotated across three rounds (one for each taxonomy), with a 14-day gap between rounds. The order in which annotators received the taxonomies was randomised to minimise ordering effects. For each round, annotators were provided with category definitions and decision rules taken directly from the original sources of each taxonomy.

This was a multi-label classification task, allowing annotators to assign multiple categories to a single advertisement where appropriate. In addition to specific manipulation categories, both "None" and "Other" were treated as valid labels.

Annotation was conducted using the *Doccano* [25] annotation platform, an open-source tool selected due to its support for multi-label text classification. Annotators were all volunteers, and details on their demographic information can be found in the appendix.

Annotators received the following instructions, alongside a description of the taxonomy they were annotating that week (they received the descriptions one at a time).

Instructions

1. Read the following description of the following taxonomy.
2. You will need to label 36-42 adverts, according to the rules of this taxonomy.
3. There is a label for *Other.* This is for when you think the advert might be manipulative, but the manipulation is not adequately described by any of the labels of the taxonomy.
4. If you think it is NOT manipulative, use the *None* label. Do not use any other labels alongside the *None* label.
5. You may label an advert with MULTIPLE labels (with the exception of the *None* label)

4.2.1 Annotator Agreement

To enable the calculation of inter-annotator agreement, a subset of advertisements was annotated by 15 annotators (excluding the author). Due to one annotator being unable to complete two assigned items, the final overlap set consisted of 73 advertisements rather than the intended 75.

Each annotator labelled 5 advertisements that were also assigned to another annotator, creating controlled overlap across the dataset. These 5 advertisements of overlap for each pair were selected randomly. This was a pragmatic compromise to balance annotator workload and dataset coverage. Increasing overlap would likely have strengthened agreement estimates, but would have reduced the total number of unique advertisements that could be annotated by the participants.

The author was excluded from the inter-annotator agreement calculations in order to minimise potential bias. The author had prior knowledge of the research objectives, taxonomy design, and the identities of other annotators. Therefore, including their annotations in agreement metrics may have artificially inflated consistency or introduced systematic bias.

This decision introduces a trade-off between maximising annotator participation and preserving the validity of agreement measurements. Nevertheless, prioritising independence in agreement evaluation was considered more appropriate given the subjective nature of the task.

4.3 LLM Annotation Setup

The following models were compared in this project: GPT-4o-mini, gemini-3-flash-preview and claude-haiku-4-5-20251001. The models were provided with the same taxonomy descriptions as the human annotators. They were also given the same annotation instructions, with the exception of the number of advertisements to label, as each model annotated the full set of 600 advertisements.

All models performed the task as a multi-label classification problem, selecting from the same set of labels available to human annotators, including the *None* and *Other* categories.

4.3.1 Few-shot Prompting

For the few-shot condition, a subset of 12 advertisements was selected for each taxonomy. These examples were chosen from advertisements with the highest level of agreement between two human annotators. For each advertisement, gold-standard annotations were constructed by taking the intersection of labels assigned by the annotators. Together the advertisements and their annotations were provided to the models as examples within the prompt.

4.4 Evaluation Procedure

4.4.1 Inter-Annotator Agreement

Inter-annotator agreement was assessed using Krippendorff's α [22] and Gwet's AC1 [23]. Krippendorff's α was computed using the *NLTK* [26] implementation, which supports multi-label annotation tasks.

Given the limitations of α in multi-label settings, Gwet's AC1 was also calculated to provide a more robust measure of agreement. Two variants of AC1 were used: (1) exact-set agreement, where annotations were treated as complete label sets, and (2) per-label agreement, where each label was treated as a binary decision and scores were averaged across labels.

Krippendorff's α captures agreement at the level of full label sets, whereas per-label AC1 captures partial agreement across individual labels. Using both metrics provides a more comprehensive view of annotator agreement.

4.4.2 Taxonomy Quality Attributes

Comprehensiveness

The feasible metric proposed for assessing comprehensiveness is the count of unclassified objects. This is defined as the number of "[objects that could not be classified] when the taxonomy is applied for its intended purpose and within scope."[9]

In our setting, annotators were instructed to use the *Other* label when an advert did not meet the criteria of any of the taxonomy's labels. We will utilise the percentage of adverts which were labelled *Other* as a marker of comprehensiveness. To convert this to a score between 0 and 1, the results are expressed as decimals and their complements are taken (i.e., $1 - p_{\text{Other}}$).

Robustness

According to A Compendium and evaluation of taxonomy quality attributes'[9] robustness can be calculated using word embeddings to compute pairwise similarities, and by counting the occurrence of *intruder* concepts: concepts that are more similar to elements outside their group than to those within it. The overall robustness score is defined as:

$$R(T) = \frac{1}{n_{\text{groups}}} \sum_{i=1}^{n_{\text{groups}}} \left(1 - \frac{n_{ic}}{n_{gc}(n_{ac} - n_{gc})}\right)$$

where n_{ic} is the number of intruder concepts in group i, n_{gc} is the number of concepts in that group, and n_{ac} is the total number of concepts in the taxonomy.

However, this formulation relies on an explicitly hierarchical taxonomy structure. The taxonomies considered in this paper are *flat*, consisting of mutually exclusive categories with no tree structure or grouping. As a result, the notion of intruder concepts within groups is not directly applicable.

Accordingly, an adaptation of the published robustness measure is used based on pairwise similarity between concepts as a proxy. Concept representations are obtained using Sentence-BERT embeddings, computed from the concept name and definition. For each concept, we identify its most similar neighbour based on cosine similarity. Robustness is then defined as one minus the average similarity to the nearest neighbour:

$$R = 1 - \frac{1}{N} \sum_{i=1}^{N} \max_{j \neq i} \ \text{sim}(c_i, c_j)$$

where N is the number of concepts in the taxonomy, and $\text{sim}(c_i, c_j)$ denotes the cosine similarity between concepts c_i and c_j.

Lower similarity between nearest neighbours indicates that categories are more semantically separated, and thus the taxonomy is more robust.

Conciseness

The most feasible conciseness measure for this paper from the list of attributes described in 'A compendium and evaluation of taxonomy quality attributes' is the number of unused categories or characteristics.

In this work, we treat each annotator as a separate instance of taxonomy application. For each annotator, we record whether a given label was used at least once across the set of annotated adverts. We then compute, for each label, the percentage of annotators who did not use that label. Higher values therefore indicate labels that are rarely used in practice.

To obtain a single conciseness score per taxonomy, we compute the mean proportion of unused labels and take its complement:

$$\text{Conciseness} = 1 - \frac{1}{M}\sum_{j=1}^{M} p_{\text{unused},j}$$

where M is the number of labels and $p_{\text{unused},j}$ is the proportion of annotators who did not use label j. Higher values therefore indicate more concise taxonomies with fewer redundant labels.

Reliability

Krippendorff's α was chosen as the metric to measure inter-annotator and intra-model agreement. This is due to its use in literature in similar tasks.[17]

For the analysis of taxonomy quality metrics, intra-model reliability was used as a proxy to intra-annotator reliability, comparing the repeated outputs of the same model (GPT-4o-mini). The model was prompted twice under identical conditions, and agreement between these outputs was calculated. This approach was adopted as it was not feasible to obtain repeated annotations from human annotators. While this does not directly reflect human intra-annotator behaviour, it nevertheless reports consistency within a single annotating agent under controlled conditions.

Mutual Exclusiveness

Mutual exclusiveness, according to 'A Compendium and evaluation of taxonomy quality attributes'[9], can be measured by the count the number of ambiguous objects: objects that are assigned to multiple categories.

To explore the concept of ambiguity, this paper will present co-occurrence matrices that visualise which labels are most frequently assigned together. These provide insight into patterns of overlap between categories.

However, while these findings are reported for completeness, mutual exclusiveness is not included in the final quality scoring. This is due to the inherent ambiguity in interpreting co-occurrence patterns. High co-occurrence between labels may reflect difficulty in distinguishing between similar categories, but it may also indicate that multiple manipulative techniques genuinely often co-occur within the same advert. As these explanations cannot be disentangled in the current setting, this metric cannot be reliably interpreted as a measure of taxonomy quality.

4.5 Requirements

Functional requirements

This project was designed to support the following functional requirements:

- Support multi-label annotation of political advertisements.
- Implement and compare three manipulation taxonomies.
- Generate LLM classifications under zero-shot and few-shot prompting.
- Compute inter-annotator and inter-LLM agreement metrics.
- Compute taxonomy quality metrics for comparative evaluation.

Non-functional requirements

The study was additionally designed to satisfy the following non-functional requirements:

- **Reproducibility:** achieved through fixed random seed and controlled prompts.
- **Consistency:** Achieved through identical instructions provided to all humans and LLMs.
- **Usability:** The use of Doccano as an annotation interface was designed to support volunteer annotators.

- **Validity / bias mitigation:** The author was excluded from inter-annotator agreement metrics and survey results.

Chapter 5

Results

5.1 Human Evaluation

5.1.1 Label Distributions

Across the annotated dataset, a substantial proportion of adverts were identified with at least one manipulative label. Depending on taxonomy, between 80.1% and 72.3% of adverts were found to be manipulative. Specifically, Simon's taxonomy returned 72.3%, Noggle's 78.7% and IMT2 80.1%.

The tables below present the distribution of labels for each taxonomy, including the categories *None* and *Other*. The *None* label was used for adverts in which annotators did not identify any manipulative techniques; *Other* captures instances where annotators identified manipulation that they felt was not adequately represented by the given taxonomy's label options.

Simon's Taxonomy Label Distribution	**(%)**
None	27.7
Playing the servant role	21.1
Guilt-tripping	17
Lying by omission	14.8
Lying by distortion	13.5
Projecting the blame	13.8
Seduction	13.3
Other	11.1
Playing the victim role	11
Brandishing anger	9.28
Rationalisation	8.43
Covert Intimidation	8.09
Shaming	7.59
Vilifying the victim	7.25
Diversion	6.41
Feigning innocence	6.07
Minimisation	5.73
Selective Inattention	3.88
Evasion	2.53
Denial	0.843
Feigning ignorance or confusion	0.675

Table 5.1: Label distribution for Simon's taxonomy

Noggle's Taxonomy Label Distribution	**(%)**
Playing on the Emotions	37
Misdirection / Misemphasis	27.3
None	21.3
Social Proof	21.6
Charm Offensive	14.1
Peer Pressure	13.9
Emotional Blackmail	12.7
Exploiting Reciprocity	10.7
Nagging	9.38
Conditional Flattery	9.38
Exploiting Conflict Aversion	8.54
Gaslighting	7.87
Other	4.36
Game Switching	4.36
Negging	1.17

Table 5.2: Label distribution for Noggle's taxonomy

IMT2 Taxonomy Label Distribution	**(%)**
Quantity violation	45.7
Manner violation	29
Quality violation	23.3
Relation violation	20.6
None	19.9
Other	12.9

Table 5.3: Label distribution for the IMT2 taxonomy

5.1.2 RQ1: To what extent do the different taxonomies align with human perceptions of manipulation in political advertisements?

Evaluating Alignment via "Other" Annotations

When considering only adverts that were perceived to be manipulative (not labelled *None*), Noggle's taxonomy exhibits the lowest percentage of *Other* labels (5.5%). In contrast, Simon (15.4%) and IMT2 (16.1%) show similar rates of *Other* usage.

Taxonomy	**Other(%)**
Noggle	5.5
Simon	15.4
IMT2	16.1

Table 5.4: Percentage of "Other" classifications among cases identified as manipulative

Annotator Preferences

Annotator responses indicate a preference for Noggle's taxonomy in multiple dimensions, with the exception of the time required to complete the annotation. Notably, zero annotators selected Noggle's as the taxonomy in which they had the least confidence.

Question	**Noggle**	**Simon**	**IMT2**
Easiest to use	10	1	4
Best suited to task	7	3	5
Best number of labels	9	0	6
Least time required	4	1	10
Least confidence in answers	0	11	4

Table 5.5: Annotator survey responses comparing perceived usability of the three taxonomies

5.1.3 Inter-Annotator Agreement

Across all taxonomies, Krippendorff's α values are below 0.67, indicating low exact-set agreement.

In contrast, AC1 (per-label) values are substantially higher. Simon shows the highest agreement (0.780), followed by Noggle (0.723), while IMT2 has a lower value (0.425). The per-label AC1 metric reflects agreement at the level of individual labels rather than the exact set used by each annotator.

AC1 (exact-set) scores remain low across all taxonomies, consistent with the pattern observed for α.

Taxonomy	**Krippendorff's α**	**AC1 (per-label)**	**AC1 (exact-set)**
Noggle	0.051	0.723	0.078
Simon	0.035	0.780	0.068
IMT2	0.020	0.425	0.090

Table 5.6: Human inter-annotator agreement across taxonomies.

5.2 LLM Results

5.2.1 RQ2. How does taxonomy choice affect LLM performance in manipulation classification?

We compare the micro and macro precision, recall and F1 scores of three different models (GPT-4o-mini, gemini-3-flash-preview and claude-haiku-4-5-20251001) tasked with annotating the dataset according to each taxonomy.

IMT2 yields the overall highest macro precision, micro F1 and macro F1 when implemented by the Claude model. IMT2 also yields the highest Micro Recall and Macro Recall when implemented by the GPT model.

To provide an overall comparison across taxonomies, we compare the Micro F1 scores averaged across models. IMT2 achieves the highest average Micro F1 (0.408), followed by Noggle (0.372), with Simon performing the worst (0.249).

Model	Micro P	Micro R	Micro F1	Macro P	Macro R	Macro F1
IMT2						
Claude	0.386	0.550	**0.454**	**0.314**	0.430	**0.336**
Gemini	0.333	0.330	0.331	0.287	0.315	0.267
GPT	0.359	**0.562**	0.438	0.300	**0.455**	0.318
Noggle						
Claude	0.320	0.455	0.375	0.219	0.248	0.171
Gemini	**0.393**	0.359	0.375	0.272	0.189	0.173
GPT	0.361	0.371	0.366	0.246	0.228	0.215
Simon						
Claude	0.234	0.252	0.243	0.147	0.181	0.129
Gemini	0.298	0.254	0.274	0.183	0.180	0.160
GPT	0.243	0.219	0.231	0.146	0.177	0.143

Table 5.7: LLM performance across taxonomies, grouped by taxonomy

Few-shot prompting

We test whether few-shot prompting improves model performance using GPT-4o-mini. The results indicate that few-shot prompting does not consistently improve performance in this setting; on the contrary, zero-shot configurations generally achieve higher overall scores across the metrics.

Setting	Micro P	Micro R	Micro F1	Macro P	Macro R	Macro F1
IMT2						
Few-shot	**0.374**	0.469	0.416	0.299	0.378	0.302
Zero-shot	0.359	**0.562**	**0.438**	**0.300**	**0.455**	**0.318**
Noggle						
Few-shot	0.361	0.334	0.347	**0.295**	0.196	0.184
Zero-shot	0.361	**0.371**	**0.366**	0.246	**0.228**	**0.215**
Simon						
Few-shot	**0.294**	0.202	**0.239**	**0.152**	0.153	0.138
Zero-shot	0.243	**0.219**	0.231	0.146	**0.177**	**0.143**

Table 5.8: GPT zero-shot vs few-shot performance, grouped by taxonomy

5.2.2 RQ3. How consistent are predictions across models for each taxonomy?

We compute the same agreement metrics used for human inter-annotator agreement, applied pairwise across models. Krippendorff's α values are consistently low. The highest agreement under this metric is observed by the GPT-Gemini pair under Noggle's taxonomy (0.203), while the lowest is for GPT-Gemini on IMT2 (-0.063).

In contrast, AC1 (per-label) scores are substantially higher. Across both the Simon and Noggle taxonomies, all model pairs achieve relatively high per-label AC1 values. The strongest alignment is observed for GPT-Gemini pair under Noggle's taxonomy (0.834), followed by Gemini-Claude under Simon's taxonomy (0.796). In comparison, IMT2 shows more variation and lower agreement, with the weakest alignment again between GPT vs. Gemini (0.109).

AC1 (exact-set) scores are low across all taxonomies, ranging. The highest exact-set agreement is observed for GPT-Gemini under Noggle's taxonomy (0.292), while the lowest is for GPT-Gemini on IMT2 (0.026).

Model Pair	Krippendorff's α	AC1 (per-label)	AC1 (exact-set)
Noggle			
Gemini vs Claude	-0.020	0.794	0.105
GPT vs Claude	0.020	0.758	0.107
GPT vs Gemini	0.203	0.834	0.292
Simon			
Gemini vs Claude	0.146	0.796	0.219
GPT vs Claude	0.078	0.765	0.113
GPT vs Gemini	0.080	0.781	0.139
IMT2			
Gemini vs Claude	0.022	0.455	0.148
GPT vs Claude	0.053	0.525	0.133
GPT vs Gemini	-0.063	0.109	0.026

Table 5.9: Pairwise agreement between LLMs across taxonomies.

5.2.3 Error Analysis

All three LLMs never used the full set of labels for any taxonomy (where we include the labels *Other* and *None).* This is most noticeable for Simon's taxonomy, where between 5 and 7 labels are never predicted, depending on the model. In particular, several labels that were frequently used by human annotators are consistently absent from model outputs, including *Playing the servant role* (21.1%) and *Playing the victim role* (11%).

Similarly, in annotating according to Noggle's taxonomy, the label *Playing on the emotions* was never predicted by any model, whilst it was the most frequently used label by human annotators at 37%.

Models also consistently differ from humans in the number of labels assigned per advert. Gemini exhibits a slight tendency to under-assign in comparison to human labels; Claude exhibits slight over-assignment; GPT displays both.

Model	Mean Human Labels	Mean Model Labels
IMT2		
Claude	1.51	2.15
Gemini	1.51	1.50
GPT	1.51	2.36
Noggle		
Claude	2.08	2.11
Gemini	2.08	1.18
GPT	2.08	1.35
Simon		
Claude	2.13	2.30
Gemini	2.13	1.82
GPT	2.13	1.92

Table 5.10: Average number of labels assigned per advert by humans and zero-shot models, grouped by taxonomy. Mean Difference is computed as model labels minus human labels.

5.3 RQ4: How do the taxonomies compare in terms of structural quality according to established taxonomy evaluation metrics?

We now present the results of the selected taxonomy quality metrics, following the framework of Unterkalmsteiner and Abdeen [9]. All measures are scaled between 0 and 1 to enable direct comparison.

5.3.1 Comprehensiveness

Comprehensiveness scores for each taxonomy are shown in Table 5.11.

Taxonomy	Other (%)	Comprehensiveness Score
Noggle	5.5%	0.945
Simon	15.4%	0.846
IMT2	16.1%	0.839

Table 5.11: Comprehensiveness scores computed as $1 - p_{\text{Other}}$.

5.3.2 Robustness

Semantic robustness scores are shown in Table 5.12. The scores show relatively little variation across the taxonomies, with IMT2 achieving the highest robustness.

Taxonomy	Number of Concepts	Semantic Robustness
IMT2	4	0.250
Noggle	13	0.227
Simon	19	0.204

Table 5.12: Semantic robustness of each taxonomy, measured as the inverse of the average similarity to the nearest neighbouring concept.

5.3.3 Conciseness

Conciseness scores for each taxonomy are shown in Table 5.13.

Taxonomy	Conciseness Score
IMT2	0.984
Noggle	0.796
Simon	0.684

Table 5.13: Conciseness scores for each taxonomy. Higher values indicate fewer unused labels and therefore greater conciseness.

5.3.4 Reliability

Both inter-annotator and intra-annotator reliability were evaluated using Krippendorff's α.

Inter-annotator reliability

Taxonomy	**Krippendorff's α**
Noggle	0.051
Simon	0.035
IMT2	0.020

Table 5.14: Human inter-annotator agreement measured using Krippendorff's α.

Inter-annotator agreement is low across all taxonomies, though Noggle performs slightly better than the others, although all results are substantially below 0.67, indicating negligible agreement amongst annotators.

Intra-Model reliability

Taxonomy	**Krippendorff's α**
Noggle	0.418
Simon	0.227
IMT2	0.278

Table 5.15: Intra-model agreement for GPT (v1 vs v2).

In intra-model reliability, Noggle again achieves the highest score.

5.3.5 Mutual Exclusiveness

Mutual exclusiveness was explored through label co-occurrence patterns. Given the multi-label nature of the task, adverts were frequently assigned multiple labels, resulting in a high number of overlapping classifications.

To examine these patterns, co-occurrence matrices are presented below.

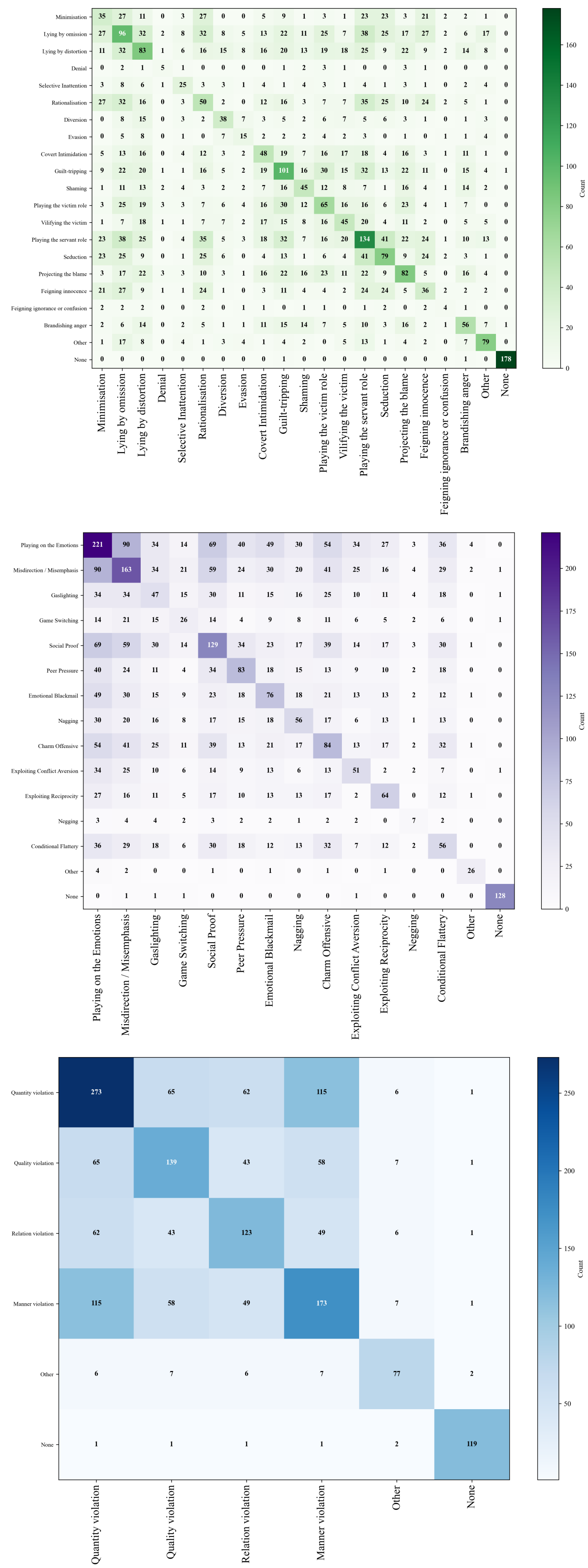

Figure 5.1: Co-occurrence matrices

These visualisations highlight patterns of overlap between categories. However, as co-occurrence may reflect either conceptual ambiguity or genuine co-occurrence of manipulation techniques, this metric is not included in the final quality scoring.

5.3.6 Overall Taxonomy Quality Comparison

The radar chart below provides a comparative overview of taxonomy quality across the following dimensions: comprehensiveness, robustness, conciseness, inter-annotator reliability, and intra-model reliability.

Noggle achieves the strongest performance in comprehensiveness and intra-model reliability; IMT2 performs most strongly in conciseness and robustness; Simon occupies an intermediate position across dimensions.

Inter-annotator reliability, measured using Krippendorff's α remains low across all three taxonomies.

Chapter 6

Discussion

6.1 Human-LLM gap in taxonomy application

The results indicate a disparity between human annotators and language models in their application of the taxonomies. Human annotators appear to show a preference for Noggle's taxonomy: the only taxonomy to receive zero votes for "least confidence". The higher confidence in Noggle arguably aligns with the low usage of the *Other* label (5.5%); annotators were less likely to encounter manipulative strategies that they felt sat outside Noggle's defined categories.

In contrast, LLMs achieve their strongest performance when applying IMT2. Across models, IMT2 attains the highest average Micro F1 score (0.408), and when implemented by Claude, also yields the highest micro recall, micro F1, macro precision, and macro F1.

One possible explanation for this lies in the structure and background of the taxonomies themselves. IMT2 takes its labels from the four Gricean maxims[14], framing manipulation as violations of conversational principles. These categories may be easier to identify by LLMs, as they may align more explicitly with structural and linguistic patterns in natural language. Furthermore, IMT2's higher semantic robustness score (0.250) may indicate a greater level of distinctness that is necessary for models to distinguish between the labels. However, it should be noted that robustness scores in this context are likely influenced by taxonomy cardinality; with only 4 categories, IMT2 naturally maintains higher semantic distance between constructs than the other two taxonomies with 13 and 19 labels.

Noggle's taxonomy, on the other hand, defines manipulation as "inducing mistakes" into an individual's reasoning process[2]. This could be argued to require reasoning about intent and psychological impact. It is possible that these features align closely with human intuition, but are less easily implemented by current LLMs.

Another factor influencing taxonomy performance may be label granularity. Simon's taxonomy, which contains 19 labels, identified 72.3% of adverts as manipulative, whereas IMT2, with only 4 categories, identified 80.1%. This could suggest that broader taxonomies may lower the threshold for classification, making it easier for instances to be labelled as manipulative. In contrast, the more specific categories of Simon's taxonomy may carry a higher burden of proof. This could result in annotators defaulting to *None* more frequently when a specific match cannot be found.

6.2 Model behaviour and agreement

Inter-LLM scores reveal very low agreement among models. Krippendorff's α values are consistently low across all pairs, indicating low agreement at the level of full label sets. The highest agreement under this metric was observed for the GPT-Gemini pair using Noggle's taxonomy (0.203); the lowest occurred for the same pair under IMT2 (-0.063). Here, a negative value indicates systematic disagreement.

In contrast, per-label AC1 scores are substantially higher. Similarly to the human results, this suggests that models somewhat agree on individual labels, though they do not often produce identical label sets. Under both Noggle's and Simon's taxonomies, all model pairs achieve relatively high per-label agreement (ranging approximately from 0.75 to 0.83).

However, IMT2 does not follow this pattern. Under the AC1 metric, agreement was moderate to low and far more variable depending on model pairs, with a range between 0.109-0.455. In particular, the GPT-Gemini pair exhibits poor per-label agreement (0.109), suggesting systematic disagreement in how the models interpret the taxonomy.

An important tension thus emerges in the results. Although IMT2 produces the strongest predictive performance, it also exhibits the weakest agreement between models. This reveals that stronger performance does not necessarily correlate with consistent performance across models.

The findings also suggest that agreement is dependent on both taxonomy choice and model pairing, indicating that taxonomy design may interact with model-specific behaviour in non-trivial ways.

Importantly, the discrepancy between near-zero Krippendorff's α and substantially higher per-label AC1 scores suggests that exact-set agreement may underestimate meaningful overlap in this multi-label setting. Given the extremely low α values, this result is particularly important for interpreting the validity of the annotation framework. The results reflect disagreement in exact label combinations, with partial consensus on manipulation labels.

6.3 Prompting effects

Few-shot prompting did not consistently improve model performance. While slight gains were observed for Simon's taxonomy (specifically in micro precision, F1, and macro precision), zero-shot configurations generally yielded higher overall scores across most metrics.

One possible hypothesis for this is that few-shot examples introduce a narrow representative bias. By presenting a subset of annotations, the prompt may encourage the model to over-rely on the labels present in the examples, rather than applying the taxonomies' definitions broadly.

In our experiments, the models were provided with 12 examples of human-annotated advertisements selected from instances with the highest inter-annotator agreement. These examples may not have captured the diversity of the broader dataset. The results may therefore suggest that a significantly larger or more diverse set of exemplars may be required to improve performance in few-shot settings.

6.4 Implications for automated classification of manipulation

The findings suggest that none of the evaluated taxonomies fully capture manipulation in a way that is suitable for both human annotation and automated classification. In particular, the presence of the *Other* label across all taxonomies indicates that each framework fails, to some extent, to account for the full range of manipulative strategies identified by annotators.

This highlights a potential limitation in current taxonomy design.

The observed disagreement between models, including two negative agreement scores (Gemini-Claude and GPT-Gemini), further indicates that LLMs cannot yet be considered reliable or objective arbiters of manipulation. Instead, their ability to apply the taxonomies appears to be sensitive to model-specific factors.

6.5 Patterns in manipulation across advertisements

Beyond evaluating the taxonomies themselves, the annotation data also reveals several patterns in the types of manipulative techniques observed across the advertisements.

A notable pattern is the prevalence of emotional manipulation. In Noggle's taxonomy, *Playing on the Emotions* was the most frequently assigned label (37%), while *Guilt-tripping* was also fairly common in Simon's taxonomy (17%). This suggests that emotional appeals may be a prominent feature of manipulation in this dataset. This observation is particularly notable given that *Playing on the Emotions* was never predicted by any of the evaluated LLMs: this may indicate a significant blind spot in current model detection capabilities.

The results also reveal that overtly hostile forms of manipulation appear less frequently than other forms. Labels such as *Brandishing anger* (9.28%) and *Covert Intimidation* (8.09%) occur less frequently than techniques such as *Charm Offensive* (14.1%) and *Social Proof* (21.6%). This may suggest that manipulation in this dataset relies more on emotionally reinforcing tactics than on explicit threats or aggression. This aligns with the acknowledgement in 'MentalManip: A Dataset For Fine-grained Analysis of Mental Manipulation in Conversations'[6] that identifying manipulation is a more complex task than detecting overtly toxic content. The results here suggest that, in this domain, this may be the case because adverts identified as manipulative are often framed as superficially positive or helpful.

Finally, the prevalence of multi-label annotations, together with low exact-set agreement between annotators, suggests that manipulation is often perceived not as a single isolated technique but as a combination of overlapping strategies. This suggests manipulation in political advertising should be viewed as layered, with multiple techniques potentially operating simultaneously within a single advert.

6.6 Future Work

These findings are derived from a single dataset of adverts and may therefore be domain-specific. Future work could investigate whether the observed trade-off between human interpretability and model applicability holds across alternative datasets, or whether it is specific to the characteristics of this corpus.

Additionally, future work could explore alternative evaluation strategies. The discrepancy between low Krippendorff's α and high per-label AC1 scores indicates that exact-set agreement may not adequately capture meaningful overlap between annotations. Developing or adopting metrics that better reflect partial agreement could provide a more accurate assessment of both human and model performance.

Prompting strategies also require more rigorous exploration. Future work could investigate the impact of example diversity, counterfactual examples, or structured prompting techniques such as chain-of-thought reasoning.

6.7 Limitations

Several limitations should be considered when interpreting these results.

- **Dataset specificity and scale:** Because annotation depended on volunteer participation, the dataset used in this study was relatively small. The findings are also derived from a subset of political advertisements from the 2022 U.S. general election. As such, observed patterns may be influenced by the political and cultural context of this dataset and should not be assumed to generalise to other domains or forms of political advertising.
- **Subjectivity of annotation:** The classification of manipulation is inherently subjective, and the low inter-annotator agreement scores indicates that different annotators often disagreed about which techniques were present. While this subjectivity is itself part of what the study seeks to measure, it also limits the extent to which any annotation set can be treated as ground truth.
- **Sparse annotation design:** Each human annotator labelled only a subset of advertisements rather than the full dataset. Although this enabled broader coverage, it reduced

the amount of repeated human annotations available for agreement analysis to a smaller subset of 73.

- **Potential confounding from taxonomy size:** Robustness and conciseness may be partially influenced by the differing number of categories in each taxonomy.

- **Dependence on selected taxonomies:** This study evaluates only three existing manipulation taxonomies. Other existing taxonomies may capture different aspects of manipulation and could yield different conclusions regarding both human agreement and LLM performance.

- **Limits of LLM evaluation:** The model results are based on a specific set of prompts, models and model versions at a particular point in time. Performance may change under alternative prompting strategies, newer model releases, or fine-tuned classifiers. The LLM findings should not be interpreted as fixed upper bounds on automated manipulation detection.

- **Adapted robustness score:** The taxonomy quality attribute robustness required adaptation to fit flat taxonomies. While these adaptations provide useful proxies, they may not capture the full intended meaning of the original measures and should therefore be interpreted cautiously.

Chapter 7

Legal, Social, Ethical and Professional Issues

This project involved the analysis of manipulative techniques in political advertising and required the participation of human annotators. As such, several ethical, legal, social, and professional considerations arise.

7.1 Ethical Considerations

A key ethical concern is the potential exposure of annotators to sensitive or polarising content. Political advertisements may address contentious topics that could cause discomfort. In order to address and mitigate this, annotators were informed in advance about the nature of the data. Participation in the annotation task was also entirely voluntary, and participants were free to withdraw at any time without consequence.

The task of identifying manipulation is inherently subjective, and annotators may be influenced by personal beliefs, introducing potential bias into the dataset. Rather than attempting to eliminate subjectivity entirely, it is accounted for by employing multiple annotators and explicitly measuring inter-annotator agreement. Variation in annotation is treated as a meaningful outcome of the work, rather than a flaw.

A further ethical consideration is the potential misuse of the findings. While this work aims to improve understanding of taxonomy utility in manipulation detection, it is possible these insights could be used to design more effective manipulative content. Although this risk is acknowledged, it is argued that the benefits of improving detection and oversight outweigh the potential for misuse.

7.2 Legal Considerations

The dataset consists of publicly available political advertisements sourced from the 'Comparable 2022 General Election Advertising Datasets from Meta and Google', and is used solely for research purposes. No personally identifiable information was collected from the annotators and all responses were anonymised. The project therefore complies with the UK General Data Protection Regulation.

7.3 Social Considerations

This work contributes to the broader understanding of manipulation in political advertising. By evaluating how different taxonomies capture manipulation, the project may inform the development of tools for identifying potentially harmful persuasive practices.

However, in manipulation classification there is a risk of over-classification, where legitimate or acceptable persuasive techniques are incorrectly labelled as manipulative. Classification results should therefore not be interpreted as definitive indicators of manipulation.

7.4 Professional Considerations

This project has been conducted in accordance with the British Computer Society (BCS) Code of Conduct and Code of Good Practice. In particular:

- **Public Interest:** The project aims to contribute to greater transparency and understanding of manipulation in political advertising.

- **Professional Competence and Integrity:** Established methodologies and evaluation metrics have been applied, and results are reported clearly and transparently.
- **Respect for Privacy:** Annotator data was anonymised and handled in accordance with data protection principles.

These principles have guided both the design and execution of the project.

Chapter 8

Conclusion

This project set out to evaluate how different taxonomies of manipulation affect the results of both human annotators and large language models when applied to a dataset of political advertisements from the 'Comparable 2022 General Election Advertising Datasets from Meta and Google'.

The results demonstrate that taxonomy design influences both human interpretation and model performance. Noggle's taxonomy appears to align most closely with human annotator behaviour, resulting in higher annotator confidence and reduced reliance on the *Other* category. In contrast, LLMs perform best under IMT2. No single taxonomy emerged as optimal across both human and automated settings.

Furthermore, the findings point towards a trade-off between human interpretability and LLM implementation. Noggle's, grounded in psychological reasoning, may better capture how humans perceive manipulation, while IMT2's structurally defined violation principles are more effectively implemented by language models.

This work contributes a comparative evaluation of three widely used manipulation taxonomies across both human annotation and LLM-based classification. By applying these frameworks to 600 political advertisements from 'Comparable 2022 General Election Advertising Datasets from Meta and Google', it provides empirical insight into how taxonomy choice shapes outcomes in manipulation detection.

Overall, this work suggests that taxonomy selection is not a neutral decision, but a central factor that influences both the consistency of human annotations and the performance of classification models. The findings indicate that progress in automated manipulation detection may depend not only on improving models, but also on critically evaluating the conceptual frameworks used to define the task itself.

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